# A phase-preserving two-branch estimate for the joint recoil Barnes limit

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**Abstract**

An exact reflection identity separates the problematic Barnes contour into a negative vertical branch and a positive-weight residue branch, with exponentially small connectors in the joint regime. The Bessel equation centers the quadratic phase exactly on both branches. This gives an absolute error of order $K^3/\nu^2 + e^{-cK}$, measured against the sum of Bessel branch magnitudes, uniformly for bounded $\alpha/K$. The phase $e^{i(K^2+\alpha^2-1)/\nu}$ is retained even when it is large. The result concerns a symmetric internal chiral graph after the fixed-mass pure-recoil limit. It is a self-reviewed proof draft, not an identification of a finite-$G$ physical correlator.

## 1 Statement and scope

Use radius one, $K, \nu > 0$, $u = v = 0$, $\omega = i\alpha$, $M = \nu^2/(2K)$. The fixed-mass $\kappa \to \infty$ limit has already been taken. No external state, operator algebra or physical normalization is newly specified here. Let $a_\pm = (1 \pm \alpha)/2$, $b = \frac{1}{2} - i(M - \nu)$, $c = \frac{1}{2} - i(M + \nu)$, and

$$Q_\nu(t) = e^{-i\pi t} \frac{\Gamma(t - i\nu/2)^4}{\Gamma(-i\nu/2)^4} \frac{\Gamma(b - t)\Gamma(c)}{\Gamma(b)\Gamma(c + t)} \Gamma(a_+ - t)\Gamma(a_- - t).$$

The contour $C$ goes upward from $-N - i\infty$ to $-N + i$, then right; $N$ is an integer with $N > \alpha/2 + 1/2$. Write

$$\mathcal{I}_\nu = \frac{1}{2\pi i} \int_C Q_\nu(t)\, \mathrm{d}t, \qquad J = -e^{i\pi\alpha} A_K - i\pi A_I,$$

$$A_K = K K_\alpha(K), \qquad A_I = K I_\alpha(K), \qquad \mathcal{E} = A_K + \pi A_I.$$

For real negative $\alpha$, use evenness and replace it by $|\alpha|$ throughout the positive scales.

**Theorem 1.** *For each finite $Y \geq 0$, there are positive constants $K_Y, \rho_Y, C_Y, c_Y$ such that, for*

$$K \geq K_Y, \qquad 0 \leq \alpha \leq YK, \qquad \rho := K^3/\nu^2 \leq \rho_Y,$$

*one has, with $S = K^2 + \alpha^2 - 1$,*

$$\left| \mathcal{I}_\nu - e^{iS/\nu} J \right| \leq C_Y (\rho + e^{-c_Y K}) \mathcal{E}. \tag{1}$$

*In particular the envelope-normalized error tends to zero when $K \to \infty$ and $\rho \to 0$, with no assumption that $K^2/\nu \to 0$.*

This includes zeros of $J$, but supplies a relative error only on regions where $|J| \geq \epsilon\mathcal{E}$. The constants are not numerically certified thresholds. Equation (1) is a leading joint estimate, not a uniform version of all previously computed fixed-parameter coefficients.

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## 2 An exact analytic splitting

Introduce

$$G(t) = (K/2)^{2t}\Gamma(a_+ - t)\Gamma(a_- - t),$$
$$V(t) = e^{-i\pi t}(K/2)^{2t}\frac{\Gamma(a_+ - t)}{\Gamma(\frac{1}{2} + t + \alpha/2)},$$
$$R_\nu(t) = e^{i\pi t}(K/2)^{-2t}\frac{\Gamma(t - i\nu/2)^4}{\Gamma(-i\nu/2)^4}\frac{\Gamma(b-t)\Gamma(c)}{\Gamma(b)\Gamma(c+t)}.$$

Thus $Q_\nu = R_\nu e^{-2\pi i t}G$. Reflection gives the exact identity

$$e^{-2\pi i t}G(t) = -e^{i\pi\alpha}G(t) + 2\pi e^{i\pi\alpha/2}V(t). \tag{2}$$

Indeed

$$e^{-2\pi i t} + e^{i\pi\alpha} = 2e^{-i\pi(t-\alpha/2)}\cos\pi(t+\alpha/2), \qquad \Gamma(a_- - t)\cos\pi(t+\alpha/2) = \frac{\pi}{\Gamma(\frac{1}{2}+t+\alpha/2)}.$$

This removes one pole family from the residue branch without discarding its phase. Integer $\alpha$ causes no singular connection coefficient in (2).

The preceding endpoint estimate implies, on every growing rectangle $|t| \le CK$, after canceling the common small Gamma factors,

$$R_\nu(t) = e^{iA(t)/\nu}(1 + \epsilon_\nu(t)), \quad A(t) = 4(t^2 - t), \quad |\epsilon_\nu(t)| \le C_C\rho e^{C_C\rho}. \tag{3}$$

It follows from sectorial Stirling and Taylor expansion of shifts $O(K)$, not a fixed-shift formula. The large Gamma arguments remain in a uniform safe sector since $K/\nu \to 0$. The quotient is analytic before reaching the poles at height $\nu/2$ or $-(M-\nu)$. On $|\Re t|, |\Im t| \le CK$,

$$|R_\nu(t)| \le \exp[C_C\rho + C_C K^2/\nu] = e^{o(K)}$$

uniformly for bounded small $\rho$. On the real axis its leading factor has unit modulus. We never extend the exact $R_\nu G$ vertical integral through its distant poles.

## 3 Finite contours and exponentially small connectors

Put

$$L = \tfrac{1}{2}\sqrt{K^2 + \alpha^2}, \quad d = \tfrac{1}{2} - L, \quad N = \lceil L \rceil, \quad r_0 = 8 + 2Y.$$

Choose $r \in [r_0K, r_0K + 1]$ at distance at least $1/8$ from both real pole lattices $a_\pm + \mathbb{Z}$. Such a choice exists: the two forbidden sets occupy at most half of a unit interval. Fix a sufficiently large constant $H = H_Y$ and set $T = HK$. Increase $K_Y$ if necessary so that

$$T < \nu/2, \qquad T < M - \nu, \qquad N > \alpha/2 + 1/2.$$

The old positive-tail lemma and optimized negative vertical estimate then truncate $C$ to

$$C_0: \quad A = -N - iT \;\longrightarrow\; B = -N + i \;\longrightarrow\; C_r = r + i$$

with error $O_Y(e^{-cK}\mathcal{E})$. Here the old positive-tail hypothesis $\nu^2 \ge 32[\max(2, Y^2/2) + 1]K^3$ is included in $\rho_Y$.

For $R_\nu G$, replace $C_0$ by

$$A \longrightarrow d - iT \longrightarrow d + iT \longrightarrow r + iT \longrightarrow C_r.$$

No pole is crossed. The small-Gamma poles lie on the real axis to the right of $d$, the numerator poles lie at height $\nu/2 > T$, and the $b-t$ poles lie below $-T$.

For $R_\nu V$, append the two sides $C_r \to r - iT \to A$. The resulting rectangle is clockwise. Its only poles are $t_n = a_+ + n < r$, $n \ge 0$, with

$$\operatorname*{Res}_{t=t_n}(R_\nu V) = -e^{-i\pi a_+} w_n R_\nu(t_n), \qquad w_n = \frac{(K/2)^{1+\alpha+2n}}{n!\Gamma(1+\alpha+n)} > 0.$$

There are no residues from the distant $b-t$ family. Keeping the signs of the clockwise contour gives

$$\mathcal{I}_\nu = -\frac{e^{i\pi\alpha}}{2\pi}\int_{-T}^{T} R_\nu(d+is)G(d+is)\,\mathrm{d}s - 2\pi i \sum_{t_n<r} w_n R_\nu(t_n) + O_Y(e^{-cK}\mathcal{E}). \tag{4}$$

Here are bounds for every connector in this construction. At $t = x \pm iHK$, $-N \le x \le r$, sectorial Stirling gives

$$\log|G(t)| \le K\{-\pi H + C_Y[1+\log(1+H)]\} + O_Y(\log K).$$

The powers of $K$ cancel between $(K/2)^{2t}$ and the two Gamma factors. For $H$ larger than twice the bounded scaled real arguments, the sector and $C_Y$ can be chosen independently of $H$. Choosing $H$ large makes the right side less than $-3K$. Multiplication by $R_\nu = e^{o(K)}$ and integration over length $O_Y(K)$ preserve exponential smallness relative to $\mathcal{E} \ge c_Y\sqrt{K}e^{-K}$.

On the right vertical side, reflection and the pole gap give

$$|G(r)| \le C_Y e^{-Kh_y(r/K)}, \qquad h_y(q) = (q+y/2)\log[2(q+y/2)] + (q-y/2)\log[2(q-y/2)] - 2q.$$

Moreover $|G(r+is)| \le |G(r)|$: shift the real Gamma arguments to positive values and use $|\Gamma(a+is)| \le \Gamma(a)$ together with the finite recurrence factors. Since $h_y(q) \ge 2q[\log(2q)-1]$, this side is small even after the multiplier and its length are included. Finally (2) bounds $|V| \le C|G|$ on the lower horizontal side and on $-T \le \Im t \le 1$. This also covers the short bottom segment from $-N - iT$ to $d - iT$. Thus no central absolute-norm estimate on the original height-one contour is used.

## 4 The positive residue branch

The standard Bessel series [2] gives

$$2\sum_{n=0}^{\infty} w_n = A_I.$$

Let $\mathcal{D} = (K/2)\partial_K$, differentiating at fixed $\alpha$. It acts on $w_n$ by multiplication by $t_n$. The Bessel equation gives $4(\mathcal{D}^2 - \mathcal{D})A_I = SA_I$. With the normalized positive weights $p_n = 2w_n/A_I$, this implies the exact centered moments

$$\mathbb{E}[A(t_n) - S] = 0, \qquad \mathbb{E}[(A(t_n)-S)^2] = 8K^2\mathbb{E}[t_n]. \tag{5}$$

For the second identity, commute $4(\mathcal{D}^2-\mathcal{D})$ through $S$, using $\mathcal{D}S = \mathcal{D}^2 S = K^2$. Also the series shift gives

$$\mathbb{E}[n(n+\alpha)] = K^2/4, \qquad \mathbb{E}[n] \le K/2, \qquad \mathbb{E}[t_n] \le (1+\alpha+K)/2.$$

Consequently the second moment in (5) is $O_Y(K^3)$. Because $A(t_n) - S$ is real, $|e^{ix} - 1 - ix| \le x^2/2$ and the first moment cancels exactly. Therefore

$$\left| 2\sum_{n=0}^{\infty} w_n e^{iA(t_n)/\nu} - e^{iS/\nu} A_I \right| \le C_Y \rho A_I. \tag{6}$$

To use (3), truncate only the exact multiplier sum at $t_n < r$. The omitted *auxiliary* weights have ratio

$$\frac{w_{n+1}}{w_n} = \frac{K^2}{4(n+1)(n+\alpha+1)}$$

uniformly bounded below one once $t_n \geq r_0 K$. Stirling at the first omitted index gives a bound $C_Y e^{-cK}\mathcal{E}$; polynomially weighted tails obey the same type of bound. Thus the second term of (4) equals $-i\pi e^{iS/\nu}A_I + O_Y(\rho A_I + e^{-cK}\mathcal{E})$. No infinite sum of the exact $R_\nu$ is invoked.

## 5 The negative vertical branch

On $t = d + is$, the two Gamma real parts are $A_\pm = L \pm \alpha/2$, both between $c_Y K$ and $C_Y K$. Real Stirling gives

$$G(d) = 2\pi e^{-KF(y)+O_Y(K^{-1})}, \qquad F(y) = \sqrt{1+y^2} - y\,\operatorname{arsinh} y.$$

The exact Gamma product identity, for $a > 0$, is

$$\log \frac{|\Gamma(a+is)|}{\Gamma(a)} = -\frac{1}{2}\sum_{j=0}^{\infty} \log\left(1 + \frac{s^2}{(a+j)^2}\right).$$

In particular, for fixed $H$ and $|s| \leq HK$,

$$|G(d+is)| \leq G(d)e^{-c_{Y,H}s^2/K}. \tag{7}$$

For example, use $\log(1+z) \geq z/(1+z)$ and retain the first $O(K)$ terms of the sum. The full product also bounds the $|s| > HK$ tail and any fixed polynomial moment exponentially in $K$. Bound its logarithm by the integral of the decreasing summand; its negative derivative is bounded away from zero for $|s| \geq HK$. These estimates and the positive $K_\alpha$ integral give

$$\frac{1}{2\pi}\int_{\mathbb{R}} |G(d+is)|\,\mathrm{d}s \leq C_Y A_K,$$

$$\int_{|s|>HK} (1+|s|^m)|G(d+is)|\,\mathrm{d}s \leq C_{Y,H,m}e^{-cK}A_K,$$

after absorbing fixed powers of $K$ into a smaller exponent.

The Mellin–Barnes formula [3] supplies

$$\frac{1}{2\pi}\int_{\mathbb{R}} G(d+is)\,\mathrm{d}s = A_K.$$

Derivatives may be taken at fixed $\alpha$ and a locally fixed vertical line, before moving it back to $d$. Its exponential tails justify differentiation. The same Bessel equation therefore gives

$$\frac{1}{2\pi}\int_{\mathbb{R}} [A(d+is) - S]G(d+is)\,\mathrm{d}s = 0. \tag{8}$$

This is an exact complex moment, not a positivity assertion. Our choice of $d$ is crucial:

$$q(s) := A(d+is) - S = -4s^2 - 8iLs.$$

The elementary complex Taylor remainder obeys

$$|e^{iq/\nu} - 1 - iq/\nu| \leq \frac{|q|^2}{2\nu^2}e^{8L|s|/\nu}.$$

On $|s| \le HK$, complete the square in (7):

$$\frac{1}{2\pi}\int_{-HK}^{HK} |G(d+is)|\,|q(s)|^2 e^{8L|s|/\nu}\,\mathrm{d}s \le C_Y K^3 A_K e^{C_Y\rho}.$$

Gaussian moments give the power $K^3$, while the displaced center costs $L^2K/\nu^2 = O_Y(\rho)$. Using (8) and its auxiliary polynomial tails,

$$\frac{1}{2\pi}\int_{-T}^{T} e^{iA(d+is)/\nu} G(d+is)\,\mathrm{d}s = e^{iS/\nu}A_K + O_Y[(\rho + e^{-cK})A_K].$$

Equation (3) contributes the same $O_Y(\rho A_K)$ after integration, because its leading modulus is $e^{8Ls/\nu}$. Combining this with (6) and the connector bounds proves Theorem 1.

## 6 Mass window and limits of the result

Since $K = \nu^2/(2M)$ and $\rho = \nu^4/(8M^3)$, the sufficient joint limit is

$$\nu \to \infty, \qquad \nu^{4/3} \ll M \ll \nu^2, \qquad |\alpha|/K \le Y.$$

This previously motivated window is now supported by a joint leading-error proof for the specified symmetric internal integral. It is not a finite-$\kappa$ or finite-$G$ theorem.

At $\alpha = 0$, the two leading terms are in quadrature, so $|J(K,0)| \ge \mathcal{E}(K,0)/\sqrt{2}$. Hence, once the error is small,

$$\frac{\mathcal{I}_\nu(K,\alpha)}{\mathcal{I}_\nu(K,0)} = e^{i\alpha^2/\nu}\frac{J(K,\alpha)}{J(K,0)} + O_Y\left[(\rho + e^{-cK})\frac{\mathcal{E}(K,\alpha)}{\mathcal{E}(K,0)}\right].$$

The amplitude convention $C$ cancels because it is independent of $\alpha$. This remains an internal chiral graph ratio, not a normalized physical four-point function. Near an auxiliary zero the absolute estimate does not locate exact finite-mass zeros or provide a relative error for their small lift.

## 7 Numerical evidence and reproducibility

The exact splitting, finite-contour orientation and all connectors are checked separately. Three small finite contours deliberately retain non-negligible boundary corrections; omitting them changes the normalized answer by roughly 0.00180, 0.0390, and 0.0111. The branch representation is also compared with six frozen full contour values from the fixed-$K$ study.

For the joint experiment use

$$\nu = 8K^{7/4}, \qquad \rho = (64\sqrt{K})^{-1}, \qquad K^2/\nu = K^{1/4}/8.$$

Thus $\rho \to 0$ while the unexpanded phase grows. The three bounded ratios $y = 0, 1, 2$ use $K = 8, 16, 32, 64$. Integrate the negative branch with 20/32 Gauss–Legendre nodes at 55/75 decimal digits, retain the positive residue sum up to a pole-safe $r$ near $12K$, and use $T = \min(8K, \nu/4)$. The largest $K$ is checked at 95 digits and with $T = 10K$ when permitted by the pole strip. These finite truncations are numerical diagnostics, not certified applications of a known minimal $K_Y$.

| $K$ | error/$\rho$, $y=0$ | error/$\rho$, $y=1$ | error/$\rho$, $y=2$ |
|---|---|---|---|
| 8 | 0.392828 | 1.002373 | 3.615498 |
| 16 | 0.364228 | 0.974595 | 3.665900 |
| 32 | 0.349064 | 0.959404 | 3.691384 |
| 64 | 0.341276 | 0.951441 | 3.705275 |

Here error means the full complex difference from $e^{iS/\nu}J$, divided by $\mathcal{E}$, not a relative difference dividing by $J$. An auxiliary-zero sample at $\alpha = 15.5$ is retained with the same normalization. The initial coarse connector grid missed its strict tolerance in two cases; the original code and output are preserved and the grid was refined, without relaxing that tolerance. The final 64 checks passed: 12 splitting/multiplier identities, 12 positive moments, nine vertical moments, six finite-contour checks, six legacy comparisons, 12 joint quadrature comparisons, six precision/extent checks and one auxiliary-zero refinement.

The next tasks are the transverse $u, v$ variables and their tails, followed by external energy contours and insertion normalization. The complete microscopic de Sitter dual remains open. This argument uses standard reflection, Mellin–Barnes inversion and Bessel identities; no priority claim is made.

## References


[1] NIST DLMF, Sec. 5.5, Eq. 5.8.3 and Sec. 5.11: Gamma reflection, modulus products and sectorial asymptotics.

[2] NIST DLMF, Eqs. 10.25.1–2: the modified-Bessel equation and positive $I_\alpha$ series.

[3] NIST DLMF, Eq. 10.32.13 and Eq. 10.32.9: positive-argument Mellin–Barnes and real-integral representations.

[4] N. M. Temme, Uniform Asymptotic Methods for Integrals, arXiv:1308.1547, Sec. 2.5. General contour context only; no joint theorem is imported.


Code and numerical records:
`code/recoil_bessel_joint.py`
`results/stage4/recoil-bessel-joint/diagnostics.json`
Prior endpoint proof:
`notes/stage4-recoil-bessel-endpoints.md`